# Pure Mathematics with Mechanics GCE A Level 2021 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0765 Pure Maths With Mechanics 1, June 2021, Advanced Level, one and a half hours). Answer ALL questions; each question has FOUR suggested answers A, B, C and D and only one answer must be marked (if more than one answer is marked the score for that question is zero). Formulae booklets and calculators are allowed. Section A: Pure Mathematics (questions 1 to 35); Section B: Mechanics (questions 36 to 50). Mobile phones are NOT allowed in the examination room.

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**SECTION A: PURE MATHEMATICS**

The vector equation of a line passing through the point $(1, 3, 4)$ and parallel to the vector $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ is

a) $\mathbf{r} = \mathbf{i} + 3\mathbf{j} + 4\mathbf{k} + \lambda(-4\mathbf{i} - 2\mathbf{k})$
b) $\mathbf{r} = \mathbf{i} - \mathbf{j} + 2\mathbf{k} + \lambda(-4\mathbf{i} - 2\mathbf{k})$
c) $\mathbf{r} = \mathbf{i} + 3\mathbf{j} + 4\mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$
d) $\mathbf{r} = \mathbf{i} - \mathbf{j} + 2\mathbf{k} + \lambda(\mathbf{i} + 3\mathbf{j} + 4\mathbf{k})$

**2.** A binary relation $R$ is defined on $\mathbb{N}$, the set of natural numbers, by $m\,R\,n \Leftrightarrow m + n$ is odd. $R$ is

a) Symmetric and reflexive
b) Symmetric only
c) Symmetric and transitive
d) Anti-symmetric

**3.** Given that the roots of the equation $x^2 - 14x + k = 0$ are $m$ and $n$ and that $3m = 4n$, the value of the constant $k$ is

a) $28$
b) $32$
c) $24$
d) $48$

**4.** The gradient of the tangent to the curve with parametric equations $x = \dfrac{2}{t}$ and $y = 3t$, where $t$ is a parameter, at the point where $t = 1$ is

a) $-\dfrac{2}{3}$
b) $\dfrac{2}{3}$
c) $-\dfrac{3}{2}$
d) $\dfrac{3}{2}$

**5.** The range of values of $x$ for which $|x| > 2 - |x|$ is

a) $x < -1$ or $x > 1$
b) $-2 c) $x < -2$ or $x > 2$
d) $-1

**6.** When a polynomial function $f(x)$ is divided by $x - 1$, the quotient is $2x^2 + x - 2$ and the remainder is $3$. The polynomial $f(x) =$

a) $2x^3 - x^2 - 3x + 2$
b) $2x^3 - x^2 - 3x + 5$
c) $2x^3 - x^2 - 3x + 3$
d) $2x^3 - x^2 - x + 2$

**7.** The value of $\theta$ for which $\dfrac{2}{3\sin\left(\theta - \frac{\pi}{3}\right) + 5}$ is minimum is

a) $\dfrac{5\pi}{6}$
b) $\dfrac{\pi}{6}$
c) $-\dfrac{\pi}{2}$
d) $\dfrac{\pi}{2}$

**8.** The set of values of $x$ for which $(x - 1)(x - 3)(2x + 1) > 0$ is

a) $x < \dfrac{-1}{2}$ or $x > 3$
b) $x < -3$ or $-1 c) $x < \dfrac{1}{2}$ or $1 d) $\dfrac{-1}{2}3$

**9.** Given that $y = x^{\sin x}$, where $x > 0$, $\dfrac{dy}{dx} =$

a) $(x\cos x)\, y$
b) $\left(-\cos x \ln x + \dfrac{\sin x}{x}\right) y$
c) $\left(\dfrac{\cos x}{x}\right) y$
d) $\left(\cos x \ln x + \dfrac{\sin x}{x}\right) y$

**10.** Let $\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} m & n \\ p & q \\ r & s \end{pmatrix}$. The order of the matrix product $\mathbf{AB}$ is

a) $2 \times 3$
b) $3 \times 2$
c) $3 \times 3$
d) $2 \times 2$

**11.** The number of arrangements of the letters of the word $MANNA$ in which the $A$'s are together is

a) $4$
b) $6$
c) $12$
d) $18$

**12.** $f(x) = \begin{cases} x^2 + 2, & 0 \leq x < 2 \\ 3x, & 2 \leq x \leq 4 \end{cases}$ Given that $f(x) = f(x + 4k)$, where $k \in \mathbb{Z}$, $f(15) =$

a) $45$
b) $11$
c) $4$
d) $9$

**13.** If the statements $p$ and $q$ are

$p$ : John is eating,

$q$ : John is playing,

the proposition $\sim q \rightarrow p$ is

a) If John is eating then he is not playing
b) If John is eating then he is playing
c) If John does not play then he will not eat
d) If John is not playing then he is eating

**14.** Values of $y$ for various values of $x$ are as shown in the table.

| $x$ | **0** | **1** | **2** | **3** | **4** |
|---|---|---|---|---|---|
| $y$ | 4 | 6 | 7 | 5 | 4 |

Using the trapezium rule $\int_0^4 y\,dx \approx$

a) $9$
b) $22$
c) $44$
d) $\frac{44}{3}$

**15.** The variables $x$ and $y$ are related by the law $y = a^2 b^x$. Reducing this law to linear form gives

a) $\log y = x \log b + 2 \log a$
b) $\log y = x \log b + a \log 2$
c) $\log y = 2 \log b + x \log a$
d) $\log y = b \log x + 2 \log a$

**16.** The general solution of the differential equation $\cos x \dfrac{dy}{dx} = y \sin x$ is

a) $y = \sec^2 x + k$
b) $y = \ln|\sec x| + k$
c) $\ln y = \ln|\sec x| + k$
d) $\ln y = \sec^2 x + k$

**17.** The sum $S_n$, of the first $n$ terms of a sequence, is $S_n = n(1 + 2n)\ln 2$. The $5^{\text{th}}$ term of the sequence is

a) $3\ln 2$
b) $19\ln 2$
c) $55\ln 2$
d) $5\ln 2$

**18.** Given that the equation $4x^2 - 3kx + 1 = 0$ has equal roots and $k > 0$, the value of the constant $k$ is equal to

a) $\frac{3}{2}$
b) $\frac{4}{3}$
c) $\frac{2}{3}$
d) $\frac{3}{4}$

**19.** The equation of the circle with centre at the point $(1, 3)$ and passing through the point $(2, 7)$ is

a) $(x-3)^2 + (y-1)^2 = 17$
b) $(x-1)^2 + (y-3)^2 = 17$
c) $(x-2)^2 + (y-7)^2 = 17$
d) $(x-2)^2 + (y-3)^2 = 17$

**20.** Given that $\frac{2x-3}{(x-3)(x-2)} \equiv \frac{P}{x-3} + \frac{Q}{x-2}$, the values of $P$ and $Q$ are respectively

a) $-3$ and $1$
b) $-3$ and $-1$
c) $3$ and $-1$
d) $3$ and $1$

**21.** The range of values of $x$ for which $\frac{x+3}{x} > 0$, $x \neq 0$, is

a) \$-3 b) \$0 c) $x < -3$ or $x > 0$
d) $x < 0$ or $x > 3$

**22.** The equation of a circle with end-points $(4, -2)$ and $(3, 2)$ of its diameter is

a) $(x-4)(x-3) + (y+2)(y-2) = 0$
b) $(x-4)(x+2) + (y-2)(y-2) = 0$
c) $(x-4)(x-3) + (y-2)(y-2) = 0$
d) $(x-4)(x-2) + (y+2)(y-2) = 0$

**23.** $\int_{-1}^{2} \frac{3+x}{2+x}\,dx =$

a) $\frac{3}{2}$
b) $\frac{1}{4} + \ln 4$
c) $1$
d) $3 + \ln 4$

**24.** The expansion of the function $\frac{1}{(1-3x)(1+x)}$ is valid for

a) $-\dfrac{1}{3} \leq x \leq \dfrac{1}{3}$
b) $-1 c) $-\dfrac13 d) $-3

**25.** The Cartesian equation of the curve whose parametric equations are $x - 1 = \sec\theta$ and $y + 1 = \tan\theta$ is

a) $y^2 - x^2 + 1 = 0$
b) $x^2 - y^2 - 2x - 2y - 1 = 0$
c) $y^2 + 3x^2 + 1 = 0$
d) $x^2 + y^2 + 2x + 2y + 1 = 0$

**26.** If the matrix $\begin{pmatrix} 6 & k & 2 \\ 4 & -3 & 3 \\ 2 & -k & 2 \end{pmatrix}$ is not invertible (i.e. is singular), the value of $k$ is

a) $-2$
b) $-3$
c) $2$
d) $3$

**27.** $\displaystyle\int 2x\,(x^2+3)^{\frac{3}{2}}\,dx =$

a) $\dfrac{1}{5}(x^2+3)^{\frac{5}{2}} + k$
b) $\dfrac{2}{5}\left(\dfrac{x^4}{4} + \dfrac{3x^2}{2}\right)^{\frac{5}{2}} + k$
c) $\dfrac{2}{5}(x^2+3)^{\frac{5}{2}} + k$
d) $\dfrac{5}{2}(x^2+3)^{\frac{5}{2}} + k$

**28.** A first approximation to the root of the equation $e^x + 2x - 1 = 0$ is $x = 1$. Using the Newton-Raphson method, a second approximation to the root of the equation is

a) $1 - \left(\dfrac{e+1}{e+2}\right)$
b) $1 - \left(\dfrac{e+2}{e+1}\right)$
c) $1 - \left(\dfrac{e-1}{e+2}\right)$
d) $1 + \left(\dfrac{e+1}{e+2}\right)$

**29.** The equation $x^3 + 3x - 5 = 0$ has a root lying in the open interval

a) $(2, 3)$
b) $(0, 1)$
c) $(1, 2)$
d) $(-1, 0)$

**30.** Given that $|z - 3| = 2|z + 3|$, where $z = x + iy$, the locus described by $z$ is

a) a circle
b) a line parallel to the $y$-axis

c) the $y$-axis
d) a line parallel to the $x$-axis

**31.** $e^{x \ln 5} =$

a) $5^x$
b) $\ln 5$
c) $e^{x^5}$
d) $e^5$

**32.** The approximate change in the value of $\ln x$ if $x$ changes from $10$ to $10.1$ is

a) $0.1$
b) $0.0001$
c) $0.001$
d) $0.01$

**33.** The vectors $2\mathbf{i} - q\mathbf{j} - \mathbf{k}$ and $-3\mathbf{i} - 2\mathbf{j} + q\mathbf{k}$ are perpendicular if the scalar $q$ is equal to

a) $1$
b) $-3$
c) $6$
d) $-2$

**34.** The vector perpendicular to the plane $3x - 5y + z + 7 = 0$ is

a) $3\mathbf{i} - 5\mathbf{j} + 7\mathbf{k}$
b) $3\mathbf{i} - 5\mathbf{j} + \mathbf{k}$
c) $3\mathbf{i} + \mathbf{j} + 7\mathbf{k}$
d) $3\mathbf{i} - 5\mathbf{j} - 7\mathbf{k}$

**35.** If $A = \tan^{-1} 5 + \tan^{-1}(-3)$, then $\tan A =$

a) $\dfrac{1}{2}$
b) $\dfrac{-4}{7}$
c) $\dfrac{1}{8}$
d) $\dfrac{-1}{7}$

**SECTION B: MECHANICS**

The position vector of a particle is $\mathbf{r} = \left[(t^3 + 2t^2 + 2)\mathbf{i} + (t^2 - 1)\mathbf{j}\right]$ m at time $t$ s. The velocity of the particle when $t = 1$ is

a) $(9\mathbf{i} + \mathbf{j})\ \mathrm{m\,s^{-1}}$
b) $5\mathrm{i}\ \mathrm{m\,s^{-1}}$
c) $(10\mathbf{i} + 2\mathbf{j})\ \mathrm{m\,s^{-1}}$
d) $(7\mathbf{i} + 2\mathbf{j})\ \mathrm{m\,s^{-1}}$

**37.** When the length of a spring is $60\%$ of its natural length, the thrust in the spring is $10\ \mathrm{N}$. The modulus of the spring is

a) $\frac{50}{3}$ N
b) $\frac{20}{3}$ N
c) $25$ N
d) $4$ N

**38.** A centripetal force of $72$ N causes a particle of mass $4$ kg to move in a horizontal circle of radius $2$ m. The angular speed of the particle is

a) $9\ \mathrm{rad\,s^{-1}}$
b) $3\ \mathrm{rad\,s^{-1}}$
c) $9\ \mathrm{m\,s^{-1}}$
d) $3\ \mathrm{m\,s^{-1}}$

**39.** Fig. 1 below shows four particles placed at the vertices of rectangle $ABCD$.

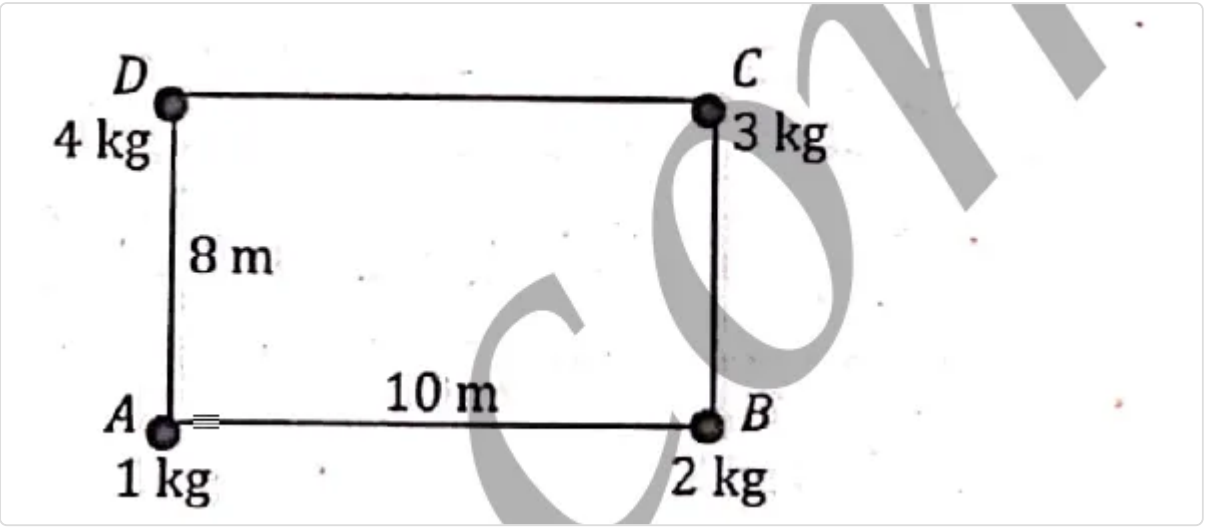



Fig. 1

The distance of the center of gravity from the side $AB$ is

a) $4$ m
b) $5.6$ m
c) $11$ m
d) $5.5$ m

**40.** A particle of mass $15$ kg rests on a smooth horizontal table. It is connected by a light inextensible string passing over a smooth pulley fixed at the edge of the table to a particle of mass $10$ kg which hangs freely. Given that the acceleration of the system is $4\ \mathrm{m\,s^{-2}}$ when it is released from rest, the force exerted by the string on the pulley is

a) $120\sqrt{2}$ N
b) $60\sqrt{2}$ N
c) $60$ N
d) $120$ N

**41.**

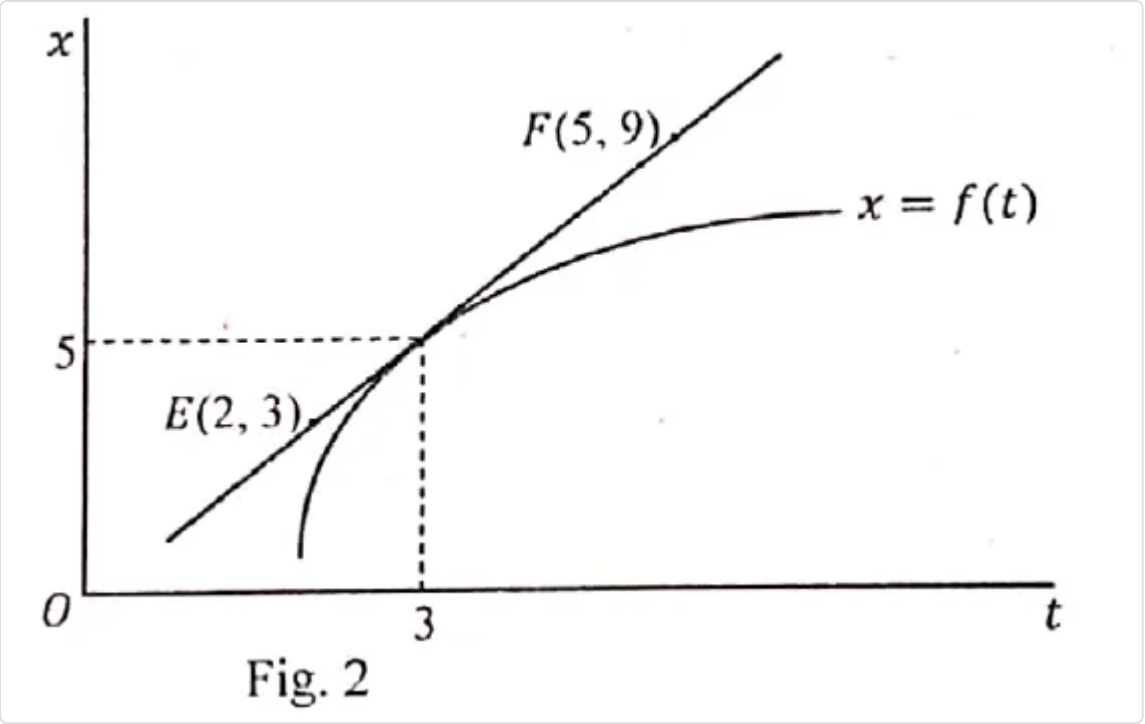


Fig. 2

Fig. 2

Fig. 2 shows a sketch of the distance-time graph of a particle whose distance $x$ m from a fixed point at time $t$ s is given by $x = f(t)$. The tangent to the curve at $t = 3$ is shown passing through two distinct points $E$ and $F$. The speed of the particle when $t = 3$ is

a) $4\ \mathrm{m\,s^{-1}}$
b) $2\ \mathrm{m\,s^{-1}}$
c) $3\ \mathrm{m\,s^{-1}}$
d) $5\ \mathrm{m\,s^{-1}}$

**42.** A uniform ladder $AB$, of length $2a$, rests in limiting equilibrium with its top end against a rough vertical wall and its lower end on a rough horizontal floor. The forces acting on the ladder are as shown in Fig. 3 below.

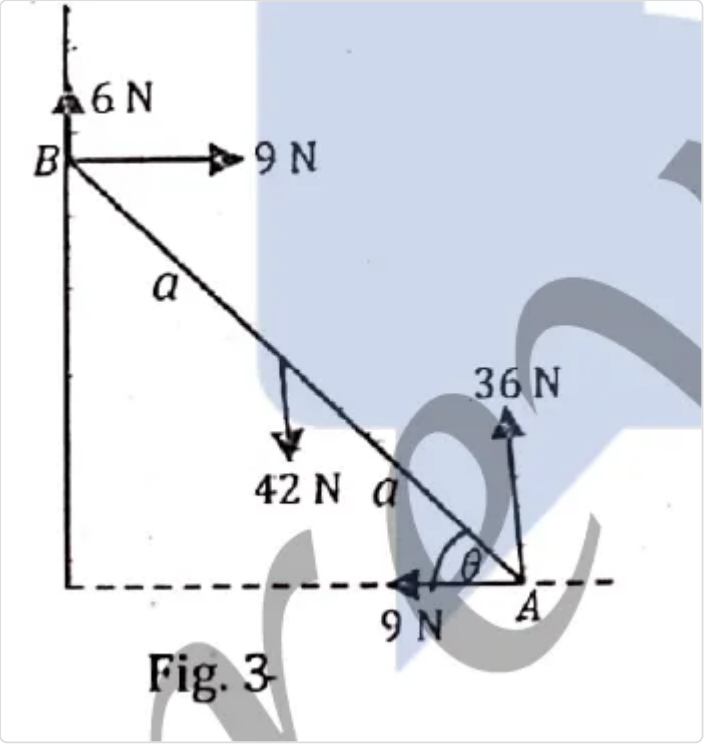


Fig. 3

Fig. 3

The angle $\theta$ which $AB$ makes with the floor is given by

a) $\tan\theta = \frac{3}{5}$
b) $\tan\theta = \frac{5}{3}$
c) $\sin\theta = \frac{3}{5}$
d) $\sin\theta = \frac{5}{3}$

**43.** The velocity of $X$ is $(4\mathbf{i} + 5\mathbf{j})\ \mathrm{m\,s^{-1}}$. Relative to $X$ the velocity of $Y$ is $(3\mathbf{i} + 2\mathbf{j})\ \mathrm{m\,s^{-1}}$ and relative to $Y$ the velocity of $Z$ is $(5\mathbf{i} - 3\mathbf{j})\ \mathrm{m\,s^{-1}}$. The true velocity of $Z$ is

a) $(12\mathbf{i} + 4\mathbf{j})\ \mathrm{m\,s^{-1}}$
b) $(9\mathbf{i} + 2\mathbf{j})\ \mathrm{m\,s^{-1}}$
c) $(7\mathbf{i} + 7\mathbf{j})\ \mathrm{m\,s^{-1}}$
d) $(8\mathbf{i} - \mathbf{j})\ \mathrm{m\,s^{-1}}$

**44.** A particle accelerating from rest has its speed $v\ \mathrm{m\,s^{-1}}$ at time $t$ s given by $v = 6t - \frac{1}{2}t^2$. The acceleration of the particle when $v = 10$ is

a) $1\ \mathrm{m\,s^{-2}}$
b) $10\ \mathrm{m\,s^{-2}}$
c) $2\ \mathrm{m\,s^{-2}}$
d) $4\ \mathrm{m\,s^{-2}}$

**45.** A smooth sphere $A$ of mass $4\ \mathrm{kg}$ travelling at $5\ \mathrm{m\,s^{-1}}$ collides directly with another smooth sphere $B$ of mass $3\ \mathrm{kg}$ travelling at $4\ \mathrm{m\,s^{-1}}$ in the opposite direction. Given that the speed of $A$ after impact is $0.5\ \mathrm{m\,s^{-1}}$, the kinetic energy of $B$ after impact is

a) $24\ \mathrm{J}$
b) $3\ \mathrm{J}$
c) $6\ \mathrm{J}$
d) $6.5\ \mathrm{J}$

**46.** The work done by a force $\mathbf{F} = (\mathbf{i} - 2\mathbf{j} - 4\mathbf{k})\ \mathrm{N}$ which moves its point of application from a point $A$ with position vector $\mathbf{r}_A = (3\mathbf{i} - 4\mathbf{k})\ \mathrm{m}$ to another point $B$ with position vector $\mathbf{r}_B = (-\mathbf{i} + 6\mathbf{k})\ \mathrm{m}$ is

a) $-24\ \mathrm{J}$
b) $-44\ \mathrm{J}$
c) $24\ \mathrm{J}$
d) $44\ \mathrm{J}$

**47.** The engine of a car of mass $1{,}200\ \mathrm{kg}$ works at a constant rate of $54\ \mathrm{kW}$ up a road inclined at an angle $\sin^{-1}\left(\frac{1}{5}\right)$ to the horizontal. Given that the non-gravitational resistance to the motion of the car is $300\ \mathrm{N}$ and taking $g$ as $10\ \mathrm{m\,s^{-2}}$, the maximum speed of the car is

a) $20\ \mathrm{m\,s^{-1}}$
b) $25\ \mathrm{m\,s^{-1}}$
c) $100\ \mathrm{m\,s^{-1}}$
d) $900\ \mathrm{m\,s^{-1}}$

**48.** A coin is biased in such a way that the probability of it landing head-up is $\frac{3}{4}$. The coin is tossed $3$ times, the probability of getting exactly one tail is

a) $\frac{9}{64}$
b) $\frac{1}{64}$
c) $\frac{27}{64}$
d) $\frac{55}{64}$

**49.** Two particles of masses $2\ \text{kg}$ and $3\ \text{kg}$ are placed at the points $(3, 1)$ and $(8, 6)$ respectively. The position of their center of mass is at the point

a) $(6, 4)$
b) $(5.5, 3.5)$
c) $(5, 3)$
d) $(4, 6)$

**50.** A force of $14\ \text{N}$ acts on a particle of mass $70\ \text{kg}$ causing the speed of the particle to increase from $3\ \text{m}\,\text{s}^{-1}$ to $7\ \text{m}\,\text{s}^{-1}$. The distance travelled by the particle during this period is

a) $40\ \text{m}$
b) $4\ \text{m}$
c) $0.3\ \text{m}$
d) $100\ \text{m}$


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